

Assignment 1 – Drone Technology Fundamentals & Components

Name: Dishant Gupta

Enrollment No.: 2407340200023

College: Rajkiya Engineering college Banda

Branch: Electrical Engineering

Year: 3rd

Mobile Number:7007201320

Email ID: dishantgupta5455@gmail.com

Internship Batch:A0826

Submission Date:31/08/2026

Q1. Drone Selection Challenge

Agovernmentagency wants to map a 500-acre agricultural field with long flight endurance, large area coverage, and accurate GPS navigation. For this requirement, I would recommend a Hybrid VTOL (Vertical Take-Off and Landing) drone.

Comparison of Drone Types

<i>Parameter</i>	<i>Multirotor</i>	<i>Fixed-Wing</i>	<i>Hybrid VTOL</i>
Flight Time	15–30 min	60–120 min	45–90 min
Coverage Area	Low	Very High	High
Hovering Capability	Yes	No	Yes
GPS Navigation	Good	Excellent	Excellent
Operational Cost	Low	Medium	High
Takeoff/Landing	Vertical	Runway Required	Vertical
Payload Capacity	Medium	Low-Medium	Medium

Technical Analysis

1. Multirotor Drone

Multirotor drones are easy to operate and can hover accurately over a location. They are suitable for inspection and photography tasks. However, their limited battery life and smaller coverage area make them inefficient for mapping a large 500-acre farm.

2. Fixed-Wing Drone

Fixed-wing drones have excellent flight endurance and can cover very large areas in a single flight. They are highly efficient for mapping missions. However, they require a runway or launcher for takeoff and landing and cannot hover over a specific location.

3. Hybrid VTOL Drone

Hybrid VTOL drones combine the advantages of multirotor and fixed-wing aircraft. They can take off and land vertically like a multirotor while achieving long flight times and large-area coverage similar to a fixed-wing drone. They also provide excellent GPS-assisted autonomous navigation.

Recommended Drone: Hybrid VTOL

For mapping a 500-acre agricultural field, the Hybrid VTOL drone is the best option because:

- Long flight endurance reduces the number of battery changes.
- Large area coverage improves mission efficiency.
- Vertical takeoff and landing eliminate runway requirements.
- High GPS accuracy ensures precise field mapping.
- Suitable for autonomous survey missions in agricultural environments.

Q2. Component Compatibility Analysis

Given Drone Configuration

<i>Component</i>	<i>Specification</i>
Frame	450 mm Quadcopter
Motor	2212 – 1000KV BLDC
Propeller	1045
Battery	3S 2200mAh 30C LiPo
ESC	30A

Why These Components Are Compatible

1. Frame and Motor Compatibility

The 450 mm frame is designed to support 2212-size brushless motors. The motor mounting holes and thrust requirements match the frame dimensions.

2. Motor and Propeller Compatibility

A 1000KV motor performs efficiently with a 1045 propeller on a 3S battery. This combination provides sufficient thrust while maintaining stable current consumption.

3. Battery and Motor Compatibility

A 3S LiPo battery supplies approximately 11.1V, which is suitable for a 1000KV motor. It delivers enough power without causing excessive motor speed.

4. ESC and Motor Compatibility

A 30A ESC can safely handle the current drawn by the 2212–1000KV motor with a 1045 propeller. This ensures reliable operation and prevents overheating.

5. Battery Discharge Capability

Battery Capacity = 2200mAh = 2.2Ah

Maximum Current = Capacity × C-Rating

=2.2 × 30

= 66A

The battery can safely deliver up to 66A, which is sufficient for the four motors and ESC system.

Scenario	Expected Problem
Smaller ESC (10A or 20A instead of 30A)	ESC may overheat, fail, or burn because it cannot handle the motor current demand.
☐ Higher KV Motor (2300KV instead of 1000KV)	☐ Excessive RPM, higher current consumption, reduced flight time, overheating, and instability.
☐ Smaller Propeller (8045)	☐ Lower thrust generation, reduced lifting capability, unstable flight.
☐ Larger Propeller (1147)	☐ Excessive motor load, overheating, higher current draw, and possible ESC damage.

Commercial Drone Used in India

Drone: DJI Matrice 350 RTK

Application: Surveying, mapping, agriculture monitoring, infrastructure inspection.

Key Specifications:

- Flight time up to approximately 55 minutes

- RTK-based high-precision positioning

- Multiple payload support

- Advanced obstacle sensing

- Long communication range

Why Suitable:

The Matrice 350 RTK provides highly accurate GPS positioning, long endurance, and reliable operation. These features make it ideal for large-scale agricultural mapping and government survey projects.